



Effects of meteorological factors on dengue incidence in Bangkok city: a model for dengue prediction

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Abstract

Dengue is of great public health concern regarding the number of people affected. In addition, climate change is associated with the recent spread of dengue fever. Effects of meteorological factors on dengue incidence from 2003 to 2019 in Bangkok city: a model for dengue prediction. Mathematical statistical applied were principal component analysis (PCA), Poisson regression model (PRM), Mann–Kendall (MK), and Sen's slope. PRM considers dengue incidence as the dependent variable and climate variables as independent variables. Meteorological factors are maximum temperature ($T_{\max}$), minimum temperature ($T_{\min}$), relative humidity (RH), and rainfall. The rainy season showed a high significant probability of occurrence for new patients. Most trends were statistically *significant* at 1% for seasonal and annual dengue cases. Another finding was that for every 5–50% of RH variation, there was an average increase (73.33–24,369.19%) in the number of dengue cases. Therefore, RH was the best predictor for increasing dengue incidence in Bangkok. In addition, predictions for dengue incidence were evaluated. This study is a significant result to warn the government, providing valuable information for human health protection.

Keywords Dengue prediction · Climate variables · Principal component analysis · Poisson regression model · Mann–Kendall

Introduction

Climate change refers to a change in the statistical weather patterns for an extended period and has significant implications for human health (Abbas and Ilyas 2018; Haines et al. 2006; Hii et al. 2012; Jácome et al. 2019; Malik et al. 2017; Patz et al. 2005; Watts et al. 2020). A study conducted by the Intergovernmental Panel on Climate Change (IPCC) showed an average temperature rise of 1.5 to 5.8 °C across the globe during the twenty-first century (IPCC 2007). There is a consensus among scientists that many dengue patients

are susceptible to climatic change (Altizer et al. 2013; Bal and Sodoudi 2020; Chevuturi et al. 2018; Relman et al. 2008; Kuhn et al. 2005; Lee et al. 2018; Parry et al. 2007; Patwardhan et al. 2018; Ramachandran et al. 2016). Dengue is a critical international public health concern leading to an increased *risk for infections* in more than 55% of the world population. Recent climate changes associated with global warming have increased the possible risk of local dengue outbreaks. It has been demonstrated that temperature, relative humidity, and precipitation changes lead to dengue disease's spatial–temporal distribution (Chen and Hsieh 2012; Phung et al. 2016; Wu et al. 2016). In addition, the temperature can affect influenza virus transmission and survival (Shaman and Kohn 2009). These studies suggest that extreme weather events in Asia have recently occurred more frequently with high intensity (Nengzouzam et al. 2019). Furthermore, strong evidence was found for the association between climatic conditions and dengue. Recently, researchers have shown an increased interest in climate and dengue; however, there is little published data in Bangkok. Therefore, this study aimed to investigate the relations between climate

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change and dengue diseases in Bangkok. The information obtained can be used to develop targeted interventions aimed at effective early warning systems for the health effects of climate change. In addition, to early warning systems, protocols are essential for information sharing, public health protection, and resource sharing and relocation, providing information for predicting future impacts.

Data collection and methodology

This study's meteorological data from the Thai Meteorological Department were $T_{\max}$, $T_{\min}$, and rainfall from 2003 to 2016. Dengue data were parallel collectors with parameters of meteorology data obtained from the Ministry of Public Health and a hospital from 2003 to 2016. Dengue data were parallel collectors with parameters of meteorology data obtained from the Ministry of Public Health and a hospital from 2003 to 2016. Linear regression, multilinear regression, PCA, MK test, and PRM were applied for data analysis. Linear trends were calculated for the monthly anomalies of temperature and rainfall. The PCA used factorial analysis to extract a small number of principal components that describe an essential function of multivariate variance. In addition, multilinear regression was employed for the relationship between climatic conditions and dengue cases. PCA transforms a set of original variables into a smaller set of linear combinations that better explain the more significant part of the original data series' variance. PCA aims to establish the factors underlying the data and determine the amount of the total variance, explained by the fewest of these factors (Preface to the First Edition 2019). MK test was used for the trend and the nonparametric, while Sen's method was applied for the magnitude of the direction. Next, a monthly lag was applied to all factors, which of these factors can directly affect the number of dengue cases in the same months. Then a correlation matrix with all the variables and their respective time lags was constructed. In this process, it was possible to identify which time lags and variables were most representative in the correlation matrix. Lastly, a PRM was utilized for meteorological variables, which affect the number of dengue cases. Then PRM was anticipated, considering the number of dengue incidences as the dependent variable and $T_{\max}$, $T_{\min}$, and Rain as independent variables. The PRM was based on the following model (de Sousa Zanotti Stagliorio Coêlho et al. 2010):

$$\text{Log}\lambda(t) = \alpha + \sum_t \beta * (X) \quad (1)$$

where $\lambda(t)$ is the dependent variable, X is the range of the independent variables, and a and b are estimated coefficients. Then data were utilized the estimated coefficients

to calculate the number of dengue incidence (NDC), the relative risk (RR) with the confidence interval (CI) using the following equations:

$$\text{NDC} = [(\exp^{(\beta * X)} - 1) * 100] \quad (2)$$

where the NDC is in percentage (%)

$$\text{RR} = \exp^{(\beta * X)} \quad (3)$$

$$\text{CI}_{95\%} = \exp[\beta \pm 1.96 * \text{se}(\beta)] \quad (4)$$

where $\text{se}(\beta)$ is the standard error of β .

PCA used factorial analysis (FA), a statistical technique designed to interpret the structure of a series of multivariate data from their respective variance–covariance matrix. The calculations of FA and PCA were carried out with SPSS software.

Results

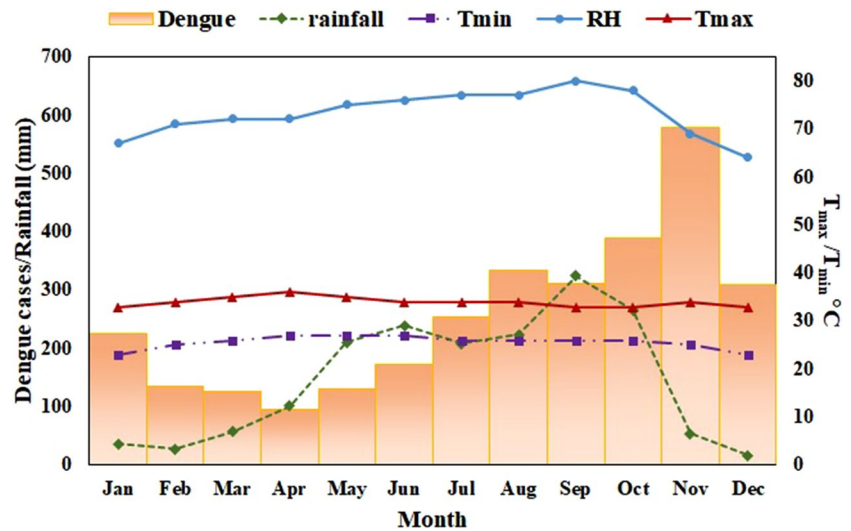
The southwest and the northeast monsoons mainly influence the climate of Thailand. According to the climate pattern and meteorological conditions, Thailand can be divided into four regions: North, Northeast, South, and Central. The central region of Thailand contains Bangkok, the capital and most populous city. Bangkok has over 5,666,264 people with a tropical savanna climate. The weather of Bangkok can be categorized into three seasons: summer, rainy, and winter, although temperatures are high year-round, ranging from an average low of 23 °C in January to an average high of 36 °C in April.

Descriptive statistics and MK test

Descriptive statistics are shown in Table 1. Dengue incidences show variability with a coefficient of variation (CV) of 42% and an average value of dengue patients of 247 people. The findings of a current study show that dengue incidences are clear seasonality (Polwiang 2015), resulting in a significant value of CV. Rainfall gives slightly higher values than other climate factors, with a mean value of 124 mm and a CV of 72%. The highest number of dengue cases was found in 2015 (1895 cases). The results obtained from the preliminary analysis of dengue and climate variables from 2003 to 2019 are shown in Figs. 1 and 2. Figures 1 and 2 show Bangkok's average monthly and annual $T_{\max}$, $T_{\min}$, rainfall, RH, and dengue cases. There has been a steady for $T_{\max}$ and $T_{\min}$; however, high values were observed from March to May, while high RH and rainfall were detected from May to October (Fig. 1). The high dengue incidences from July to December, reaching a peak in November, are

Table 1 Descriptive statistics of the variables

Variables	Average	Median	Minimum	Maximum	Std.	CV (%)
Dengue cases	247	242	104	400	109	42
T _{max} (°C)	34	34	33	36	1	3
T _{min} (°C)	26	26	23	27	1	5
Rainfall (mm)	124	130	15	268	89	72
RH (%)	73	74	64	80	5	7

Fig. 1 The average monthly T_{max}, T_{min}, RH, Rain, and dengue for 2003–2019 in Bangkok

likely related to waterlogging at the end of the rainy season, causing the spread of the disease in the early winter (Hii et al. 2012; Polwiang 2015). Between 2003 and 2019, rainfall was recorded at 601.1 mm in September 2012, and RH also showed a high value in September 2012 at 85%. High T_{max} and low T_{min} were 37.4 °C (in May 2014) and 20.8 °C (in December 2014) (Fig. 2). Figure 2 also presents most of the high dengue incidence each year coinciding with RH. On the annual timescale, the significant increasing trends in dengue incidence are observed at the 1% significance level, giving the rate of 30 people/year (Table 2). On the seasonal scale, the highest significant dengue increases are found in the rainy season at 40 people/year. These results support the findings of a great deal of the previous work by Lee et al. (2017).

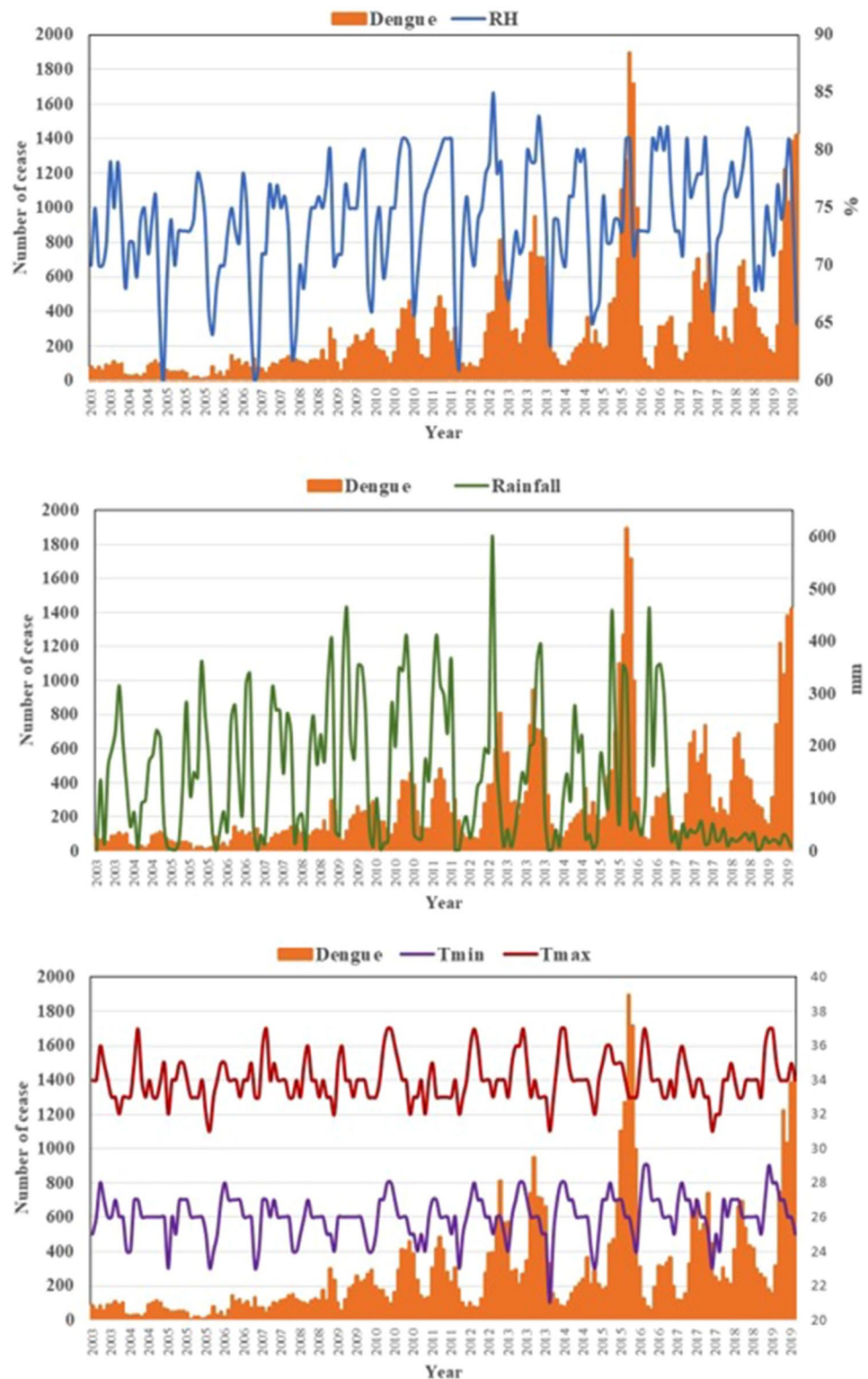
PRM model and PCA

Numerous studies have revealed the influence of climate variables on dengue incidence using PRM (Benedum et al. 2018; Hii et al. 2012; Xu et al. 2014). Table 3 presents the Pearson correlation matrix for Dengue and meteorological parameters. Those parameters with higher statistical significance and correlation coefficients were applied to PRM. Table 4 presents the highest correlation and statistical significance based on a correlation matrix. With the collinearity

between both factors, the importance of one decreases the relevance of the other when analyzed simultaneously. Thus, the selection criterion chose those variables with the highest correlation coefficient.

Meteorological variables such as T_{min}, rainfall, RH, and dengue incidence at a lag of one month were significant, while insignificant variables were rejected (T_{max}). In the model, the number of dengue incidences (1-month lag), T_{min} (7-month lag), and RH (9-month lag) were positively correlated with the dengue incidence. In contrast, rainfall (3-month lag) was negative with dengue incidence. The positive and negative association with dengue incidence was significant at a 95% confidence level. Therefore, incidence rate ratios of dengue incidence were calculated with a 95% confidence interval. In Table 4, the T_{min} at 7-month lag, the corresponding month's rainfall at 3-month lag, and RH at 2-month lag had a more significant influence on dengue infections than other variables. Then, the PRM estimates parameters for dengue, calculating the relative risk influenced by T_{min}, rainfall, and RH and the possibility of new dengue incidents. Table 5 for every 1–10 °C of T_{min} variation shows an average decrease of –18.13 to –86.472% of dengue cases. Every 10–100 mm of variation in rainfall and 5–50% of the variation in RH, there is an average decrease of –2.96 to –25.92% and an increase of 73.33 to 24,369% in the number of dengue cases, respectively.

Fig. 2 The average yearly $T_{\max}$, $T_{\min}$, RH, Rain, and dengue for 2003–2019 in Bangkok



As shown in Fig. 3, the model behaves satisfactorily, based on the dengue estimation from the PRM equation. From these results, it is possible to verify that the increasing dengue is due to an increase in RH or a decrease in $T_{\min}$ and rainfall. Furthermore, from the coefficients found in the

PRM model, it was possible to calculate the rise/fall in dengue cases. The RR is calculated to confirm how the increase/decrease in dengue incidence is related to the influence of $T_{\min}$, RH, and rainfall (Fig. 3). For every 1–10 °C of the variation in $T_{\min}$, the decrease would be from 0.82 to 0.14

Table 2 Results of the statistical tests for seasonal and annual $T_{\min}$, $T_{\max}$, and rainfall over the period 2003–2019

Stations	Test	Summer	Rainy	Winter	Annual
Bangkok	Z_s	3.38**	3.83**	3.74**	4.16**
	Q_{med}	13.22**	40.15**	38.70**	29.88**

Z_s is Mann–Kendall test, Q_{med} is Sen's slope estimator

*Statically significant at $P < 0.05$

**Statically significant at $P < 0.01$

*** Statically significant at $P < 0.001$

Table 3 Correlation between dengue, meteorological variables ($T_{\min}$, rainfall, RH)

Lag	$T_{\min}$ ($P < 0.005$)	Rainfall ($P < 0.005$)	RH ($P < 0.005$)
1	−0.2	−0.1	0.11
2	−0.1	−0.17	0.05
3	−0.02	−0.26	−0.07
4	0.10	−0.25	−0.12
5	0.18	−0.16	−0.05
6	0.28	−0.05	0.11
7	0.34	0.07	0.24
8	0.32	0.18	0.36
9	0.22	0.18	0.40
10	0.08	0.12	0.35
11	−0.07	0.02	0.24
12	−0.09	−0.13	0.12

The values in italic have higher statistical significance and correlation coefficients

Table 4 Poisson regression model and coefficients for dengue

Prediction	Incidence rate ratio	CI	P value
Intercept	645.88	540.66–792.79	< 0.001
$T_{\min}$ lag7	0.74	0.74–0.75	< 0.001
Rainfall lag3	1.00	1.00–1.00	< 0.001
RH lag9	1.10	1.10–1.10	< 0.001

The values in bold have statistical significant at $P < 0.001$

Table 5 Variation in the range of $T_{\min}$, rainfall, and RH in the numbers of dengue cases

Variables	The numbers of dengue cases (%)									
	$\nabla 1^\circ\text{C}$	$\nabla 2^\circ\text{C}$	$\nabla 3^\circ\text{C}$	$\nabla 4^\circ\text{C}$	$\nabla 5^\circ\text{C}$	$\nabla 6^\circ\text{C}$	$\nabla 7^\circ\text{C}$	$\nabla 8^\circ\text{C}$	$\nabla 9^\circ\text{C}$	$\nabla 10^\circ\text{C}$
$T_{\min}$	−18.13	−32.97	−45.12	−55.07	−63.21	−69.88	−75.34	−79.81	−83.47	−86.47
	$\nabla 10\text{ mm}$	$\nabla 20\text{ mm}$	$\nabla 30\text{ mm}$	$\nabla 40\text{ mm}$	$\nabla 50\text{ mm}$	$\nabla 60\text{ mm}$	$\nabla 70\text{ mm}$	$\nabla 80\text{ mm}$	$\nabla 90\text{ mm}$	$\nabla 100\text{ mm}$
Rainfall	−2.96	−5.82	−8.61	−11.31	−13.93	−16.47	−18.94	−21.31	−23.66	−25.92
	5%	10%	15%	20%	25%	30%	35%	40%	45%	50%
RH	73	200	420	802	1464	2611	4599	8045	14,017	24,369

with a $C/99$ 0.74:0.75. For every 10–100 mm of the variation in rainfall, the decrease would be from 0.97 to 0.74 with a $C/99$ 1.00:1.00. In contrast, for every 5–50% of RH variation, the increase would be from 1.73 to 244.69 with a $C/99$ 1.10:1.10, respectively (Table 5). Therefore, RH implies an increase in dengue cases.

The PRM model was produced from 2003 to 2019 (188 months), and the estimating prediction was performed using the same set of data. As you can be seen in Fig. 4, the predictions were addressed through observation cases. It was found that the predicted and observed values have a similar pattern, having a peak and lowest in the same month. There are 0.72 correlations with a 95% confidence interval of 0.40–1.01 with a P value less than 0.001.

PCA is employed to verify that the climate variables affect the number of dengue cases. The PCA follows the PRM, where PC1 is represented by RH, rainfall, and $T_{\min}$, while $T_{\max}$ represents PC2. Table 6 illustrates the preliminary analysis of the climate variables for the considered factors, PC1, and PC2. There 88% of the data variance (PC1 and PC2) were satisfied with the explanation of the procedure. The PC1 showing 58% of the variance is RH, demonstrating a positive association with dengue. The PC2 account for 30% of the variance in $T_{\max}$. Therefore, the sum of PC1 and PC2 should represent 88% of the original data set in a study with such an approach. One of the more significant findings from this study is that humidity is a better predictor for the dengue incidence model than other climate variables. This finding was also reported by Xu et al. (2014).

Conclusions and discussions

The relationship between climate variables and dengue in Bangkok from 2003 to 2019 was investigated using PCA, PRM, MK, and Sen's methods. Our findings reveal the climate pattern of Bangkok, showing high rainfall and RH from May to October and low rainfall from November to April. Nevertheless, temperatures are high throughout the year. The annual and seasonal dengue trends showed significant growth at the 1% significance level. According to

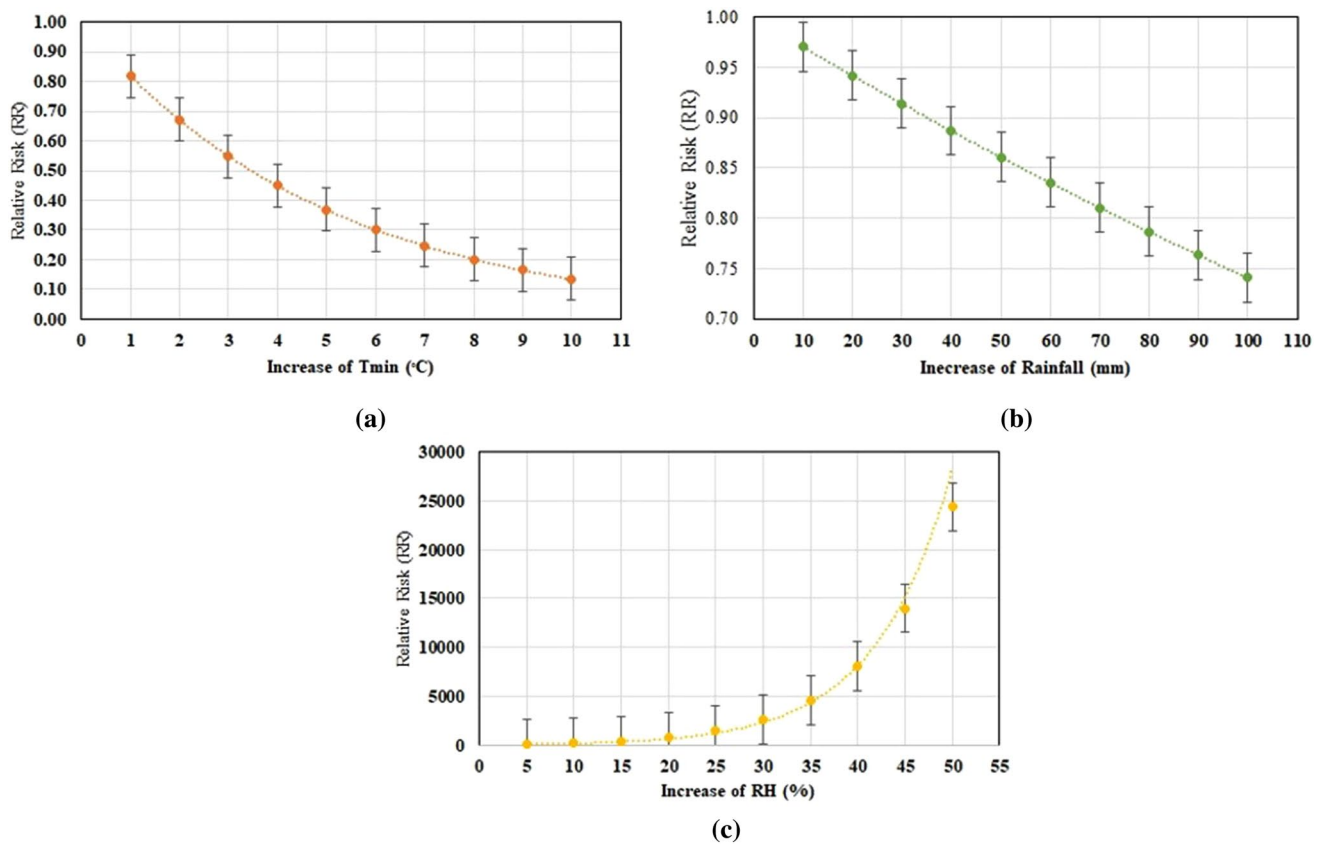
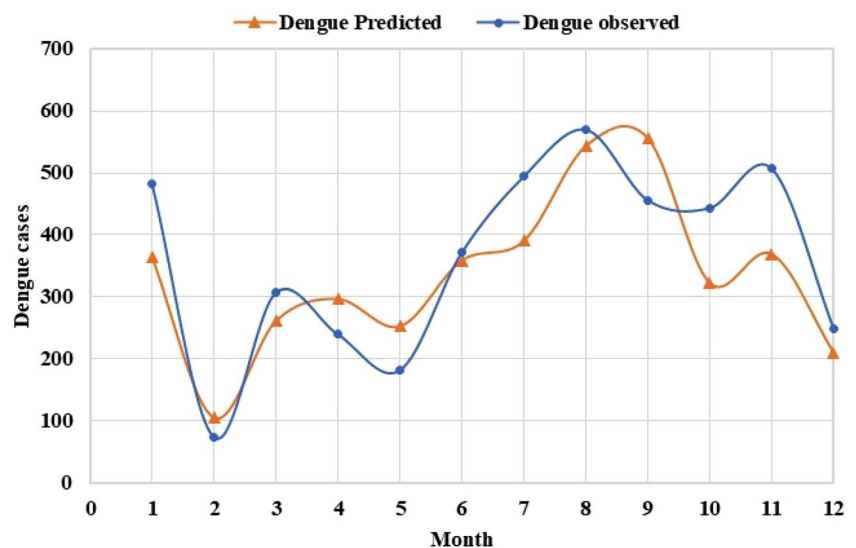


Fig. 3 The increase in the risk of occurrence of dengue cases with the influence of the **a** T_{min} and **b** rainfall **c** RH

Fig. 4 Comparison of observed (blue line) and predicted (orange line) dengue incidence using PRM from January 2003 to December 2019



the analysis, it is possible to state that dengue incidence and meteorological variables show a statistically significant correlation with dengue hospital admissions. The dengue incidence and climate variables analysis showed a positive correlation (RH) and a negative correlation (rainfall and T_{min}). Therefore, based on climatic conditions, the PRM model

is sufficient to estimate the number of dengue incidences. The findings also found that PCA gave the highest weight to the RH, followed by rainfall. Therefore, the PCA verifies the findings from the PRM model, showing that RH is the variable that most influenced the increase in dengue cases. This accords with our earlier observations, which showed

Table 6 Correlations between variables and factors: factor analysis with factor loadings and the meteorological variables

variables	PC1	PC2
$T_{\min}$	0.46	0.88
Rain	0.60	−0.48
RH	0.68	−0.08
Eigen values	1.742	0.91
% of variance	58	30

that the increase in RH revealed a consistently strong correlation with the rise in dengue (Huang et al. 2013; Meng et al. 2021). These results also reflect Karim et al. (2012) and Alkhaldy (2017), who found that RH significantly affected dengue. In addition, high rainfall comes with high RH corresponding with high dengue, resulting in the spreading of the disease.

Furthermore, the southwest monsoon from the Indian Ocean is usually across Thailand from June–September, resulting in high dengue during this period. Finally, this method can be used to monitor dengue incidences and can provide an increase in the number of cases of the disease. Hence, too abundant rain from the southwest monsoon can be a good condition for the mosquito's life cycle; however, excess water will inhibit the mosquito's life cycle, letting it "wash out" the mosquitoes. Therefore, this work is a significant step in understanding dengue for the best solution to control the number of dengue incidences. Consequently, RH is the best predictor for the number of dengue incidences in Bangkok.

On the other hand, PCA is an outstanding possibility for applying many factors to order how important each parameter is. This technique was compared to PRM, giving very satisfactory. Finally, the PRM model can estimate dengue incidence when climatic conditions change. This suggests that the finding of the PRM model can be enhanced with more parameters, providing the best prediction for dengue incidences. An implication is that this methodology can monitor dengue incidences and their increase. Our results might also help health workers predict cases and prepare to reduce hospital admissions.

Author's contribution All authors contributed to the study conception and design. Material preparation, data collection and analysis were performed by WK, OP, WN, YJ, SC. The first draft of the manuscript was written by WK and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

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Data availability The datasets generated during and/or analyzed during the current study are not publicly available due to privacy and ethical concerns, neither the data nor the source of the data can be made available but are available from the corresponding author on reasonable request.

Declarations

Conflict of interest The authors have no conflicts of interest to declare that are relevant to the content of this article.

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